PHYSICS OF SENSOR PROBLEMS

Q1. A platinum resistance thermometer has a resistance of 180 Ω at 20°C. Calculate its resistance at 60°C (α_{20}= 0.00392)

Given:- α_{20}= 0.00392, R_0=180 Ω

Formula:- $R=R_0 (1 - \propto \Delta T)$

Solution:-

$R=R_0 (1 - \propto \Delta T)$

R= 180 [100039260 °C -20°C)]

R= 180 [1000392 x 40°C)]

R= 180 [1 - 0.1568]

R=180 x 0.8432

R= 151.78 Ω

Ans:- The resistance at 60°C is 151.78 Ω

Q2. A platinum resistance thermometer has a resistance of 100 Ω at 25 C^0. Find its resistance at 50°C. The resistance temperature coefficient of platinum is 0.00392Ω /Ω°C.

If the thermometer has a resistance of 200 Ω calculate the value of temperature.

Given:- **Resistance 25 C^0 = 100 Ω**

Formula:- $R= R_0 (1 + \alpha_0 \Delta t)$

Solution:-

Step1 :Using the linear approximations, the value of resistance at any temperature

R= Ro (1 + do At)

R=100(1 + [(0.00392) X (50 - 25)°C])

= 100(1 + [(0.00392 x 25) °C])

= 109.8 2

Step2 :Suppose t_2 is the unknown temperature then

200 = 100(1+ [(0.00392) x (t- 25)°C])

2 = (1+[(0.00392) × (t_2-25) °C])

2-1 = [(0.00392) x (t_2-25)°C])

$$(t_2 - 25)°C = \frac{1}{0.00392} \quad \text{therefore, } t_2 = 280°C$$

Ans:- The value of temperature is 280°C